Chapter 4

Artifact: finite-space approximation

The approximation theorem on a finite probability space

Bühler et al., Proposition 4.3, prove the following on a finite probability space. In the proof we flag each use of finiteness of Ω as (F1) to (F3); these are exactly the points that need attention on a general probability space.

Proposition 4.1. *Suppose the probability space $\Omega = \{\omega_1, \dots, \omega_N\}$ is finite with $\mathbb{P}[\{\omega_i\}] > 0$ for all i, and $\rho : \mathbb{R}^N \to \mathbb{R}$ is a convex risk measure. Then for every claim X one has $\lim_{M\to\infty} \pi_M(X) = \pi(X)$.*

Proof. By monotonicity of (π_M) (Lemma 4.1) it suffices to show that for every $\varepsilon > 0$ some M satisfies $\pi_M(X) \le \pi(X) + \varepsilon$.

Choose $\delta \in \mathcal{H}^{\mathrm{u}}$ with

$$\rho\big(X + W(H \circ \delta)\big) \le \pi(X) + \frac{\varepsilon}{2}. \tag{4.1}$$

Since each δ_k is $\mathcal{F}_k$-measurable with $\mathcal{F}_k = \sigma(I_0, \dots, I_k)$, and since δ_{k-1} is a function of $(I_0, \dots, I_{k-1}, \delta_{k-2})$ we can recusively define f_k so that $\delta_k = f_k(I_0, \dots, I_k)$.

(F1) Integrability of the hedge ratios. Because Ω is finite, each δ_k takes finitely many values and every component of f_k lies in $L^1(\mu_k)$ where μ_k is the law of $(I_0, \dots, I_k)$. Finally, Hornik's theorem yields networks $F_{k,m} \to f_k$ in $L^1(\mu_k)$.

(F2) From L^1 to pointwise convergence. On a finite Ω, L^1 convergence implies pointwise convergence: $\delta_k^m := F_{k,m}(I_0, \dots, I_k) \to \delta_k$ for every $\omega \in \Omega$.

Set $G_m := ((H \circ \delta^m) \cdot S)_T$ and $C_m := C_T(H \circ \delta^m)$. Since H and the stochastic integral are continuous, $G_m \to G$ pointwise. Furthermore, using upper semicontinuity of the c_k and continuity of H again, one has $\limsup_m C_m = \overline{C} \le C := C_T(H \circ \delta)$ pointwise.

(F3) Continuity of the risk measure. Identifying random variables with vectors in $\mathbb{R}^N$, the convex map $\rho : \mathbb{R}^N \to \mathbb{R}$ is continuous. Continuity and monotonicity of ρ give

$$\lim_{m\to\infty} \rho(X + G_m - C_m) = \rho(X + G - \overline{C}) \le \rho(X + G - C) \le \pi(X) + \frac{\varepsilon}{2}$$

by (4.1). Pick m large enough so that the left-hand side is $\le \pi(X) + \varepsilon$. Because δ^m belongs to $\mathcal{H}_M$ for all M once m is large enough (nested network classes), $\pi_M(X) \le \pi(X) + \varepsilon$ follows. □

What breaks on a general probability space

Remark 4.2. (F1): hedge ratios need not be integrable. On general Ω, the representing functions f_k need not lie in $L^1(\mu_k)$, and then Hornik's theorem cannot even be invoked: f_k is not an element of the space in which networks are dense.

Example 4.3. Let I_0 have a standard Cauchy law μ_0 and $\mathcal{F}_0 = \sigma(I_0)$. The strategy $\delta_0 = f(I_0)$ with $f(x) = |x|$ is perfectly admissible as a measurable adapted process, but $f \notin L^1(\mu_0)$, so no sequence of μ_0-integrable functions (in particular no sequence of networks with bounded activation, which are bounded functions) converges to f in $L^1(\mu_0)$.

Remark 4.4. (F2): only subsequential almost sure convergence (harmless). On general Ω, $L^1(\mathbb{P})$-convergence yields almost sure convergence only along a subsequence. This is harmless for the argument: the proof only requires *one* good approximating strategy, so we may freely pass to subsequences.

Remark 4.5. (F3): the risk measure is not continuous along almost sure convergence. This is the decisive failure. On $\mathbb{R}^N$, convexity of ρ implies continuity. On an infinite-dimensional domain, a convex risk measure need not be continuous along almost sure convergence. The direction needed in the final step of the proof is an *upper* semicontinuity:

$$\limsup_{m\to\infty} \rho(Y_m) \le \rho(Y) \qquad \text{whenever } Y_m \to Y \ \mathbb{P}\text{-a.s.}$$

This fails even for the entropic risk measure, which satisfies the half-bounded *lower* semicontinuity (A3) of Chapter 3.

Example 4.6. Work on $\Omega = (0, 1)$ with the Borel σ-field and Lebesgue measure, let $\lambda > 0$ and let $\rho_\lambda(Y) = \frac{1}{\lambda} \log \mathbb{E}[e^{-\lambda Y}]$ be the entropic risk measure. Set $A_m := (0, e^{-m})$ and

$$Y_m := -\frac{2m}{\lambda} \mathbf{1}_{A_m}.$$

For every $\omega \in (0,1)$ one has $\omega \notin A_m$ for all $m > \log(1/\omega)$, so $Y_m \to 0$ everywhere. But

$$\mathbb{E}\big[e^{-\lambda Y_m}\big] = 1 - e^{-m} + e^{-m}e^{2m} = 1 - e^{-m} + e^{m} \longrightarrow \infty,$$

so $\rho_\lambda(Y_m) \to \infty$ while $\rho_\lambda(0) = 0$. Small events carrying large losses make the risk explode along an almost surely vanishing sequence.